RHEL ADMINISTRATION

Scheduling a Future User Job

at · atq · at -c · atrm

What deferred jobs are

The idea

Run commands once, at a specific future time. Example: a heavy maintenance task at midnight, or a safety job that resets a firewall in ten minutes unless you cancel it.

The tools

The **at** package is installed and enabled by default. It gives you the **atd** daemon plus the **at** and **atq** commands.

The queues

Any user can queue jobs. Queues run a to z and A to Z. Higher letters run with higher priority.

at TIMESPEC: queue a job

```
user@host:~$ at now +5min < myfile
warning: commands will be executed using /bin/sh
job 2 at Wed May 14 20:40:00 2025
```

now +5min run in five minutes

< myfile read the commands from a script file

job 2 the unique job number

Ctrl+D on an empty line ends manual typing

TIMESPEC: time, then date

TIMESPEC	Runs
now +5min	in 5 minutes
teatime tomorrow	16:00 tomorrow
noon +4 days	12:00, four days on
5pm august 3 2025	that exact date

Date only: time defaults to now. Time only: next time it matches.

```
user@host:~$ at 21:03 < myfile
job 3 at Wed May 14 21:03:00 2025
user@host:~$ at 21:00 < myfile
job 4 at Thu May 15 21:00:00 2025
21:00 had passed, so tomorrow
```

atq: list pending jobs

```
user@host:~$ atq
28 Mon May 19 05:13:00 2025 a user
29 Tue May 20 16:00:00 2025 h user
30 Wed May 21 12:00:00 2025 a user
```

28
unique job number

Mon May 19 05:13:00 2025
execution date and time

a
queue (default is a)

user
job owner

Inspect and remove a job

at -c JOBNUMBER

```
user@host:~$ at -c 28
(environment variables)
(commands that will run)
```

Shows the job's saved environment, copied from your shell when you queued it, then the commands.

atrm JOBNUMBER

```
user@host:~$ atrm 28
user@host:~$ atq
29 Tue May 20 16:00:00 2025 h user
```

Removes the job. Silent on success; atq confirms it is gone. Use it once the safety job is no longer needed.

Command cheat sheet

Command	What it does
at TIMESPEC < file	Queue a job from a script
atq or at -l	List your pending jobs
at -c JOBNUMBER	Show a job's environment and commands
atrm JOBNUMBER	Remove a scheduled job